

Moment/SOS relaxations for AC optimal transmission switching: consolidated findings

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Abstract

We apply sparse moment/sum-of-squares (Lasserre) relaxations to AC optimal transmission switching (AC-OTS) on the 18 PGLib-OPF “api” test cases of Taheri and Molzahn, with every branch switchable. Two threads are pursued: certified lower bounds from per-clique mixed-order relaxations, and correlated randomized rounding of the switching binaries from the relaxation’s pseudo-moments. We obtain certified gaps below 1 % on eight of the eighteen cases and below 0.5 % on five, and we find solutions that improve on the reference values for four cases. We report three findings that we believe matter beyond this benchmark: (i) certification quality, not relaxation strength, drove an earlier incorrect ranking of big-M versus no-big-M formulations; (ii) a network preprocessing step (low-impedance bus merging) silently destroyed the validity of the bounds, producing a “lower bound” 72 % above a known feasible solution, and removing it changes the certified gap by between +0.04 and −0.07 percentage points — it is free, and on the largest case an improvement; and (iii) the collapse of randomized rounding above 118 buses is caused by the *number* of lines opened simultaneously, not by network islanding, and capping it raises the AC-feasible sample rate on 300-**ieee** from 0 % to 91 % and yields two new best-known solutions, on 179-**goc** and 500-**goc**.

1 Scope

The problem is AC-OTS written as a polynomial optimization problem in rectangular voltage coordinates (see `docs/formulations.md`). Binaries $z_\ell \in \{0, 1\}$ select branch status. The benchmark is the set of 18 PGLib-OPF v21.07 “api” (congested) cases used by Taheri and Molzahn, with every branch switchable; their AC-OPF, O-DC-OTS and AC-OTS (Juniper) values serve as reference points.

Two research threads:

1. **Certified lower bounds** from sparse, per-clique mixed-order moment relaxations.
2. **Correlated randomized rounding** of the binaries from the relaxation’s pseudo-moments, with continuous recovery by Ipopt.

2 Formulation and method

Binaries by basis reduction. We impose $z \in \{0, 1\}$ by substituting $z^k \rightarrow z$ in the monomial basis rather than adding the equality $z^2 = z$. Keeping it as an equality makes the SDP feasible set lower-dimensional, Slater’s condition fails, and the order-2 interior-point solve degrades badly. This is a practical, not cosmetic, distinction.

Switching is degree three. The flow equations become $p_\ell = z_\ell P_\ell(V)$, which is cubic. Order-1 (Shor) relaxations cannot represent this, and the resulting bounds collapse to uselessness (78,116 on the 73-bus case, 0 on the 118-bus case, 20,079 on the 200-bus case).

Order policy rather than a global order. Only cliques containing a binary are raised to order 2 (`binary_clique_order`); all others remain at order 1. In practice 40–48% of cliques are order 2 and the largest PSD block has size 54–75, which is what makes the degree-3 constraints affordable at these network sizes.

Variants. `mixed` (big-M), `mixed_nobigM`, `adj16` (adjacent-clique merging, maximum size 16), `adj16_pairs` (additional order-2 blocks on binary pairs), and, from Experiment 4, `base` and `conn` (radial fixings plus connectivity cuts).

Certified bounds from inexact solves. MOSEK terminates at `SLOW_PROGRESS` on these problems, so its dual objective is *not* a valid bound. Following Oustry et al. [1], we compute a bound valid at *any* dual point θ :

$$F(\theta) = \theta_t - \sum_{\alpha} |r_{\alpha}(\theta)| \max |x^{\alpha}| + \sum_k \rho_k \min(\lambda_{\min}(A_k(\theta)), 0),$$

where all correction terms are non-positive. Four variants are implemented (box; box with tight trace bounds; implicit-Gram; per-block hybrid) plus an L-BFGS-optimized certificate on a smoothed surrogate. Because F is valid for every θ , a badly converged solve still yields a rigorous bound.

3 Main results

Costs in \$/h. “Best known” is the minimum of our best found solution and the paper’s AC-OPF, O-DC-OTS and AC-OTS values. Gap is (best known – certified bound) / best known. Every bound below is checked against the best known feasible cost; none violates it.

case	AC-OPF	O-DC-OTS	AC-OTS	ours	best known	bound	gap
3-lmbd	11,236	10,636	10,636	10,636	10,636	10,636	0.00%
5-pjm	76,377	75,190	75,190	75,190	75,190	73,376	2.41%
14-ieee	5,999	5,999	5,999	5,999	5,999	5,735	4.39%
24-ieee-rtts	134,944	122,283	119,743	119,743	119,743	111,741	6.68%
30-as	4,996	2,797	2,797	2,797	2,797	2,767	1.06%
30-ieee	18,044	17,939	17,936	17,940	17,936	17,287	3.62%
39-epri	249,672	246,850	246,723	246,561	246,561	245,369	0.48%
57-ieee	49,290	49,290	49,274	49,272	49,272	49,224	0.10%
60-c	185,239	182,028	182,028	182,028	182,028	180,836	0.65%
73-ieee-rtts	422,627	413,133	385,194	389,142	385,194	368,338	4.38%
89-pegase [†]	130,175	100,702	100,344	100,503	100,344	99,858	0.48%
118-ieee	242,237	195,918	180,312	222,376	180,312	169,523	5.98%
179-goc [†]	1,932,044	1,931,004	–	1,928,125	1,928,125	1,835,909	4.78%
200-activ	35,701	35,701	35,701	35,701	35,701	35,670	0.09%
240-pserc [†]	4,640,589	4,627,155	–	4,635,816	4,627,155	4,606,386	0.45%
300-ieee [†]	684,985	684,985	683,968	684,401	683,968	678,283	0.83%

case	AC-OPF	O-DC-OTS	AC-OTS	ours	best known	bound	gap
500-goc	692,407	692,271	–	676,183	676,183	667,141	1.34%
1354-pegase [†]	1,498,271	1,496,750	–	1,498,117	1,496,750	1,487,079	0.65%

[†] Rigorous bound on the exact (unmerged) model; see Section 5. All other rows never involved merging, so *every* bound in this table is rigorous.

Solutions improving on the reference. 179-goc 1,928,125 and 500-goc 676,183 (both new; Section 6.1), 39-epri 246,561 (opening lines {4, 6}, better than the paper’s Juniper AC-OTS 246,723); 500-goc 685,131 (better than the paper’s O-DC-OTS 692,271); 57-ieee 49,272; 300-ieee 684,401.

Where the relaxation is weak. The largest gaps are on *small*, highly combinatorial cases (24-ieee-rtts 6.68 %, 118-ieee 5.98 %, 179-goc 4.92 %, 14-ieee 4.39 %). Small does not mean easy here: a larger fraction of the network is switchable relative to its redundancy.

4 Certification matters as much as the relaxation

Re-certifying every stored relaxation with the optimized certificate tightened every case, in several substantially: 179-goc 6.28 % $\rightarrow$ 4.88 %, 60-c 1.08 % $\rightarrow$ 0.65 %, 200-activ 0.30 % $\rightarrow$ 0.09 %. Two conclusions:

- **The big-M versus no-big-M ranking was a certification artifact.** Big-M appeared inferior because its certificates were looser, not because its relaxation was weaker. Certified properly, big-M gives the *better* bound on 179-goc (1,836,683 versus 1,810,514). No-big-M remains 1.5–2 $\times$ faster to solve.
- **Tight trace bounds never helped**, because the residual term dominates the box certificate. A clean negative result.

After re-certification the certified bound lies within 0.25 % of the raw SDP objective on every relaxation, and within 0.03 % on most, so the remaining gaps are attributable to the relaxations rather than to inexact solves.

5 Rigour of the bounds

5.1 A preprocessing step invalidated the bounds

Low-impedance bus merging (`merge_zmax`, default 10^{-3}) was applied on 5 of the 18 cases. It is *not* a relaxation, for three independent reasons:

1. **Restriction (voltages).** Buses joined by a merged tie are forced to share one (e, f) pair; the original problem does not require this.
2. **Restriction (switching).** Merged ties are removed from the switchable set, i.e. forced closed. Minimizing over a *subset* of switching configurations can only raise the optimum. This affects 178 branches on 1354-pegase.

3. **Relaxation (losses).** Merged ties are modelled as lossless, which acts in the opposite direction.

Because the effects have opposite signs, the merged optimum is neither provably above nor below the true one, so a bound computed on a merged model is not a valid lower bound for the original problem.

5.2 The failure this caused

On 1354-pegase the consequence was not subtle. Merging at the default threshold merged a tie sitting exactly at its thermal limit (flow/rate = 1.000), which rendered the model *infeasible*: with every switch closed the merged POP is `LOCALLY_INFEASIBLE`, while the unmerged one solves to 1,498,271.03, exactly the paper’s AC-OPF cost. An SDP relaxation of an infeasible problem can take any value, and indeed the re-certification reported “lower bounds” of 2,580,996, 2,612,454 and 3,772,084 against a known feasible cost of 1,496,750 — exceeding it by 72% to 152%.

The mechanism was isolated by scaling only the merged ties’ thermal ratings: at $\times 1.00$ and $\times 1.02$ the model stays infeasible, at $\times 1.10$ it solves. Voltage differences across merged ties are negligible at the true solution (maximum 0.001,94 p.u.), so the shared-voltage restriction was nearly harmless; the binding thermal limit was not.

Remedies now in place. A self-validating merge rule (`safe_merge_exclusions`) compares the all-closed AC-OPF of the merged model against the unmerged one and un-merges the most heavily loaded tie until they agree, terminating in the worst case with nothing merged. On 1354-pegase it un-merges 6 of 184 ties and reproduces the reference AC-OPF to -0.018% ; on every other case it excludes nothing. A regression test (`scripts/validate_merging.jl`) checks all 18 cases, and the experiment scripts now verify that no certified bound exceeds a known feasible cost, with the tabulation step failing outright if one does.

5.3 Removing merging costs almost nothing

Since the unmerged model *is* the exact problem, and the certificate is valid at any dual point, rigour is obtained simply by not merging. Re-running all variants on the exact model:

case	merged ties	merged bound	rigorous bound	certified gap
179-goc	2	1,836,682.93	1,835,908.57	4.883% $\rightarrow$ 4.923%
89-pegase	19	99,890.19	99,858.29	0.452% $\rightarrow$ 0.484%
300-ieee	2	678,466.75	678,283.45	0.804% $\rightarrow$ 0.831%
240-pserc	55	4,606,997.21	4,606,386.34	0.436% $\rightarrow$ 0.449%
1354-pegase	178	1,486,086.12	1,487,078.55	0.712% $\rightarrow$ 0.646%

Rigour is free. Across the five affected cases the change in certified gap ranges from $+0.040$ to -0.066 percentage points: it costs almost nothing on four cases and *improves* the largest, 1354-pegase, where the exact model gives a strictly better bound than the merged one. **All eighteen cases now carry rigorous bounds** (thirteen never merged anything; five re-run on the exact model).

Two caveats. First, part of even that small deficit is budget rather than modelling: the merged numbers come from runs with a 1,800s certificate budget and the rigorous ones from 900s; in a controlled comparison at equal budget the unmerged bound was *better* on 179-goc and 300-ieee

(and five times faster on the latter). Second, merging is not obsolete: the 89-pegase big-M variant fails numerically when unmerged, returning a garbage point and certificates near -1.6×10^8 . That is still a valid (if useless) bound, and the no-big-M variants supply the best bound anyway, but `merge_zmax` cannot simply be set to zero and the merge machinery deleted.

6 Rounding at scale

On small enumerable instances (Experiments 1–2, with ground truth by enumeration) correlated schemes outperform independent rounding, confirming that order-2 pseudo-moments carry usable joint information.

At scale the picture changes sharply. Over 100 samples per scheme, the AC-feasible fraction is 39–55 % on 89-pegase and 24–37 % on 240-pserc, but 0–2 % on 118-ieee, 179-goc, 300-ieee, 500-goc and 1354-pegase — under *every* scheme, including island-guarded sampling and island repair.

The cause is cardinality, not connectivity. Independent and Gaussian rounding decide each line separately, so the number of lines opened simultaneously concentrates around $\sum_{\ell}(1 - y_{\ell})$, which is 25 on 300-ieee and 724 of 1,807 binaries on 1354-pegase. Capping it, while still letting the relaxation choose *which* lines (weighted sampling without replacement by $1 - y_{\ell}$, implemented as a Gumbel top- k draw):

case	independent	cap 8	cap 5	cap 3	cap 1	best cost found
118-ieee	0/50	0 %	6 %	22 %	58 %	225,486 (+25.05%)
300-ieee	0/30	47 %	67 %	77 %	93 %	684,766 (+0.12%)

On 300-ieee this is the first time rounding alone, with no local-search polishing, has produced competitive solutions at that size. On 118-ieee feasibility is restored but solution quality is not, and the reason is diagnostic: the relaxation implies roughly 11 openings whereas the best known solution opens 37, so *no* sampler drawing from those fixed marginals can reach it. At that size the limitation is the pseudo-distribution, not the rounding scheme.

Islanding is a partial explanation only. Island repair does not restore feasibility, but the dominant rejection reason varies by case. Of 100 independent samples, AC local infeasibility versus island-screen rejection splits 78/22 on 118-ieee, 78/18 on 200-activ and 95/5 on 500-goc, but 18/82 on 179-goc, 46/54 on 300-ieee and 0/100 on 1354-pegase. Cardinality capping helps because it addresses both at once: opening fewer lines produces fewer islands *and* less network stress.

6.1 How many lines should a sample open?

`scripts/experiment5.cardinality.jl` sweeps the cap directly, reusing the stored marginals so no SDP is re-solved. $N = 100$ ($N = 60$ on 500-goc), evaluated on the original network.

Two new best-known solutions, both verified independently by re-solving the AC-OPF for that switching configuration from scratch on the original network:

- 179-goc: 1,928,124.99, opening only two lines {147, 158} — 2,851 (0.148 %) below our previous incumbent, and below the paper’s O-DC-OTS (1,931,004) and AC-OPF (1,932,044).

case	$E[\#opened]$	independent	best cap	feasible	best cost	vs best known
118-ieee	10.9	0 %	$k=3$	17 %	225,486.25	+25.05%
179-goc	39.3	0 %	$k=2$	78 %	1,928,124.99	− 0.15 %
200-activ	19.2	5 %	$k=2$	77 %	35,700.87	+0.00%
240-pserc	33.0	26 %	$k=5$	89 %	4,638,056.84	+0.24%
300-ieee	25.0	0 %	$k=1$	91 %	684,584.26	+0.09%
500-goc	50.5	0 %	$k=6$	52 %	676,182.83	− 1.31 %
89-pegase	37.2	38 %	$k=8$	74 %	101,016.95	+0.67%

- 500-goc: 676,182.83, opening six lines $\{131, 228, 464, 607, 629, 698\}$ — 8,949 (1.306 %) below our incumbent and 2.3 % below the paper’s O-DC-OTS (692,271).

Both tighten their certified gaps, since the best known cost falls: 179-goc 4.923% $\rightarrow$ 4.783% and 500-goc 2.626% $\rightarrow$ 1.337%, roughly halving the latter.

The effect is large and consistent: independent rounding lands at 0–38 % AC-feasible, capping at 52–91 %. The useful caps are 1–8, always far below the relaxation’s own expectation, which remains useless everywhere (0–2 %), and quality is flat across neighbouring caps, so the result does not depend on tuning the cap. 118-ieee stays the exception: feasibility is restored but cost remains +25%, because its optimum opens 37 lines while the marginals imply about 11.

1354-pegase is excluded: its stored marginals come from the pre-fix merged model, which was infeasible (Section 5), so they cannot be used for rounding. A dimension guard in the script catches this; regenerating them requires re-running experiment 3 or 4 on the corrected model.

7 Negative results worth recording

- Tight trace bounds in the box certificate never changed the answer.
- Island repair and guarded sampling help only where rounding already works.
- Connectivity cuts and radial fixings improve the bound on 89-pegase and 1354-pegase but do not improve rounding on 118-ieee, 179-goc or 300-ieee.
- Capping openings at the relaxation’s own expectation is insufficient (2 % and 0 %); the useful caps are far tighter, 1–8 lines.
- The “api” cases are genuinely tight: on 118-ieee, 100 of the 186 single-line openings are AC-infeasible on their own, and feasibility is *not* monotone in the set of opened lines.

8 Open problems

1. Sweep the opening cap $k_{\max}$ and re-run Experiments 3 and 4 with cardinality-limited sampling.
2. For cases requiring large coordinated rewirings (118-ieee, 89-pegase, 73-ieee-rtts), move from resampling fixed marginals to conditioning and re-solving (randomized diving), which changes the distribution; and test conditional moments $\mathbb{E}[V \mid z]$ as Ipopt warm starts.

3. Understand why the 118-ieee marginals are so concentrated (expected 11 openings against an optimum opening 37). This is a relaxation-tightness question and connects to the relaxation-selection thread.
4. The ML-guided relaxation-selection thread remains untouched; non-ML baselines (OBBT, QC envelopes, dual- and eigenvalue-guided order elevation) should come first.

Artifacts

Julia package MomentMINLP (JuMP, MOSEK, PowerModels, Ipopt/HSL). Supporting notes: `results/writeup.md` (the Markdown counterpart of this document), `results/merging_findings.md`, `results/rounding_scale_findings.md`, `results/experiment4_findings.md`, `results/recertify_findings.md`, `results/numerics_findings.md`. Regression tests: `scripts/validate_merging.jl` (18/18) and `scripts/test_islands.jl` (96/96).

References

- [1] A. Oustry, C. D’Ambrosio, L. Liberti and M. Ruiz, “Certified and accurate SDP bounds for the ACOPF problem,” *Power Systems Computation Conference (PSCC)*, 2022.